

Exploring the Effect of Emotion Awareness Intervention on Reducing FoMO

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ABSTRACT

Fear of missing out (FoMO) refers to one's perception that others live a better life as well as one's desire to keep constantly linked to what others are doing via social media. Our study aims to develop an emotion awareness intervention based on the Satir Model to help people become more aware of their emotions after using Instagram through a set of reflection questions and guiding prompts. We designed and built a web-based application called Being to deliver these content in a 10-day field study. Our findings suggested that participants generally found the reflection questions helped them better perceive how they use Instagram in terms of their use habits and mental states. The combination of reflection questions and guiding prompts helped improve participants' emotion regulation skills and enabled them to develop strategies of using less Instagram. We conclude by discussing implications for future researchers interested in developing similar interventions.

CCS CONCEPTS

• Human-centered computing; • Human-computer interaction (HCI); • Empirical studies in HCI;

KEYWORDS

Fear of missing out, Emotional well-being, Emotion awareness, Social media

ACM Reference Format:

Shih-Chu Chen, Yen-Han Chang, Jing-Han Huang, Ching-Wen Hsu, Chih-Hsuan Lin, and Pei-Yi (Patricia) Kuo. 2022. Exploring the Effect of Emotion Awareness Intervention on Reducing FoMO. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519839>

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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00
<https://doi.org/10.1145/3491101.3519839>

1 INTRODUCTION

Social media plays a special role in our daily life [29]. The overuse or inappropriate use behavior on social media brings people negative impacts on their self-esteem, disturbed sleep patterns [23]. Prior research has pointed out several threats that over-using social media could bring to people [11, 22]. These include addiction, fear of missing out (FoMO), mood and anxiety disorders, cyberbullying, and so on [22]. FoMO is a feeling that one perceives others are living a better and fulfilling life [7]. This feeling makes people feel lonely, low confidence, and de-attached [11].

Although some research aims to help people reduce FoMO, most of them stay at the conceptual stage and/or provide static content. For instance, an information campaign raising people's understanding of FoMO was implemented in one study, and a comprehensive guidebook containing various solutions addressing FoMO was designed [2, 4]. Besides, some passive approaches dealing with social media use were proposed in prior work, such as limiting social media use time to around 30 minutes per day [14]. This could be difficult to implement in real-life context as recent statistics has shown a growing trend of people's social media use time [8].

As people's degree of social media use relates to emotion regulation (more use leads to poor emotion regulation) [15, 37], it is essential to investigate how to improve one's emotion regulation. Self-awareness is a fundamental stimulator of self-regulation, and it impacts one's emotional self-regulation [30]. Thus, our study adopts the Personal Iceberg Metaphor of Satir Model [3, 28], a theory used in psychological counseling, to help people become more aware of their emotions during and after social media use, with the ultimate goal that better emotion regulation skills and emotional well-being are realized. Specifically, we have the following two research questions: (1) RQ1: How do reflection questions focusing on raising emotion awareness reduce participants' FoMO, and improve their emotional well-being? ; (2) RQ2: How do reflection questions plus guiding prompts focusing on raising emotion awareness make participants reduce FoMO, and improve their emotional well-being?

2 RELATED WORK

2.1 Fear of Missing Out (FoMO) and Relevant Interventions

People's lives are increasingly influenced by social media. Prior studies point out that higher Facebook usage has been linked to

loneliness and low self-esteem [16, 31]. One of the common issues aroused from over-using social media is called Fear of Missing Out (FoMO). FoMO is defined as a widespread fear that others always live a better and more fulfilling life, and that the person has a strong desire to be constantly linked to what others are doing via social media [7].

Moreover, some researchers took a practical approach to design applications and information to address FoMO. For instance, an awareness campaign against FoMO and a self-help guide - FoMO reduction (FoMO-R) – offers coping skills like promoting positive thoughts and managing expectations [2, 4]. Notifications alerting people to stop using social media are also used to minimize FoMO, yet people find them a bit annoying [34]. 21 developed guidelines against FoMO adopting the co-design approach with different stakeholders (parents, schools, and the government) [21]. Another study reflects on how social networking sites can be better designed given the rise of the FoMO issue, including features to allow users to filter out unimportant information and prioritize the information based on their importance [1]. Based on these works, our study seeks to adopt a more active and individual approach to engage people in the process of emotion reflection after using social media.

2.2 Social Media, Emotion Regulation, and Emotion Awareness

Surfing on social media may be a common action taken by many adults when they face emotional distress. Prior research has suggested that people with difficulties with self-regulation may turn to social media more easily - insufficient self-regulation may lead to smartphone addiction [26]. Besides, emotional regulation refers to a person's ability to recognize, understand, accept, acquire, control, and work on emotional reactions [5, 12, 26]. People who fail to regulate emotions are subjective to emotional problems and mental disorders [25]. Since people's degree of social media use relates to emotion regulation (more use leads to poor emotion regulation) [15, 37], it is essential to investigate how to improve one's emotion regulation.

Self-awareness refers to the ability to be the focus of one's attention. In the case of social media use, self-awareness makes people actively identify, process, and save information about themselves. Self-awareness is also a fundamental stimulator of self-regulation, and it impacts one's emotional self-regulation [27, 30]. Existing efforts aiming to increase one's self-awareness include tangibilizing one's emotions using a clay kit [18], performing mindfulness practices [32, 38], using deep learning model for users to self-examine one's depression status based on the content of his or her Facebook posts [36], empowering users to create their emotion map to raise their awareness [13], and so on.

While there is a growing attention on FoMO, limited research examines the effect of these resources on people in a real-life context [2, 4]. As self-awareness plays an important role in helping people modulate their emotions during social media use, this study aims to help people reduce FoMO through engaging them in an emotion awareness intervention.

3 METHODS

3.1 Study Design and Procedure

We conducted a 10-day field experiment with a total of 20 participants. Participants were randomly assigned to the control and the experiment groups. As Instagram (IG) is one of the most popular platforms among college students, we created a public IG account and used the chatroom to interact with participants. We also designed and built an emotion awareness intervention called *Being* for our field experiment. Our study procedure consists of three stages – pre- field experiment, a 10-day field experiment, and post-field experiment (1).

3.1.1 Pre- field experiment. We first used pre-screening survey to recruit participants suffering from FoMO using the FoMO scale [6]. Participants were then randomly divided into the control and experiment groups. We ensured equal gender distribution in each group. Next, we hosted an online briefing session for our participants to introduce the experiment process and asked them to complete a pre-study survey composed of the WLEIS (Wong and Law Emotional Intelligence Scale) [17, 35] and CERQ-short (Cognitive Emotion Regulation Questionnaire - short) [9, 10].

3.1.2 10-Day field experiment. During the field experiment, we utilized an official IG account to communicate with our participants and remind them to fill out the form on the *Being* app daily (one reminder per day). Participants in both groups were asked to meditate for 30 seconds to cool down their minds before answering questions in the *Being* app. Participants in the control group need to only answer questions related to reflection on their IG use, while those in the experiment group need to follow and respond to the guiding prompts in addition to reflection questions related to IG use.

3.1.3 Post- field experiment. Participants retook the survey measuring their FoMO, emotion regulation, and emotional intelligence scales after the field experiment. We also invited participants who performed better and worse during the experiment for post-study interviews (about 30 to 60 mins) to better understand their feelings and reactions to the *Being* app and its content. We invited 10 people for interviews (5 from the control group and 5 from the experiment group).

3.2 Design and Development of the *Being* App

3.2.1 The *Being* app: System design and functions. We designed an emotion awareness intervention called *Being* using Heroku for our field experiment (Figure 2 - Left). It includes three main features: (1) provide reflection questions on IG use, (2) record current emotions, and (3) show a history of participant recordings. There are two versions of the *Being* app (one for control group and one for experiment group). The system features of both versions are identical – allowing participants to record their daily reflection by clicking “START” button and review their previous records simply through the daily record cards. The difference between two versions of the app is the content of the form, which we will explain in the following paragraph.

3.2.2 Content design of the *Being* app. Our intervention has two types of content delivered via two versions of the *Being* app. The

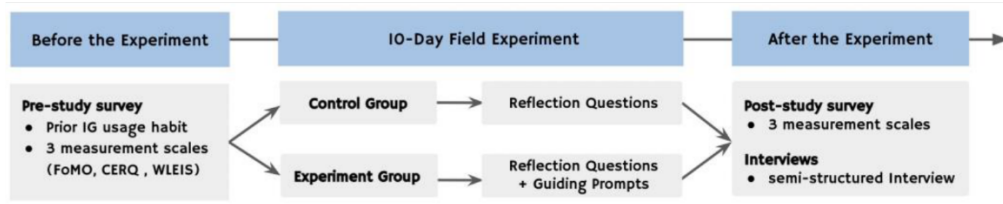


Figure 1: Study procedure

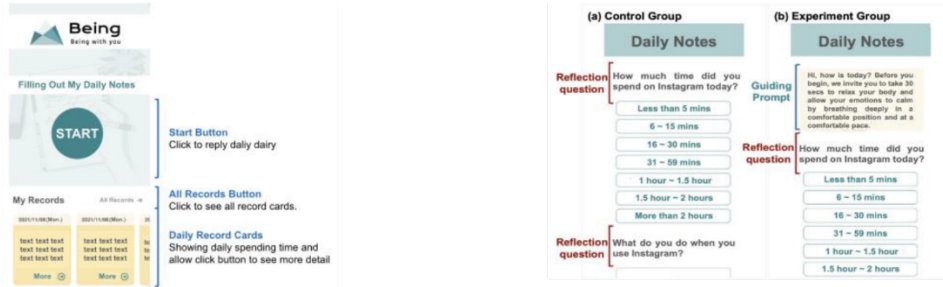
Figure 2: Daily reflection form homepage and recording screenshots in the *Being* app.

Table 1: The guidelines of content design informed by the Personal Iceberg Metaphor

Theory	Guidelines of Content Design	Reflection Questions in <i>Being</i> App
The Personal Iceberg Metaphor [28]		
Behavior and Coping	Understand the facts experienced by users.	Q1 & Q2
Feelings	Guide users to express their emotional and physical feelings .	Q3
Feelings behind feelings	Guide users to explore and aware of their deeper feelings .	Q4
Perceptions	Guide users to tell the meaning or point of view behind their emotional and physical feelings.	Q5
Expectations	Guide users to clarify their expectations for an event or their own state.	Q6
Yearnings		
Self	Guide users to appreciate themselves, increase positive energy, or express expectations of themselves.	Q7

content of these two versions is informed by the Personal Iceberg Metaphor of Satir Model [3, 19, 28]. The Satir Model theory commonly used in psychological counseling encourages people to further reflect on the emotions and consequences of their behaviors by using the metaphor of an iceberg (examining what is underwater). The reflection questions on IG use and the guiding prompts concretizing the scenario of Satir's model invited participants to think more deeply and understand the rationale and expectation behind their emotions through the narrative description. Table 1 shows the guidelines of our content design which is based on the Satir Model theory [28] and the extended 6A steps of self-awareness [19].

The content of the guidelines and questions based on the Satir model are adapted to examine emotions aroused after IG use. A pre-test with five participants was conducted before the field experiment to ensure the understanding of these reflection questions and guiding prompts, and whether the length of use time is reasonable. Table 2 shows the reflection questions for the control group (CG)

as well as the guiding prompts and reflection questions for the experiment group (EG). Except for the additional guiding prompts (from G1 to G5) and four more questions (from Q4 to Q7) prompting participants in the experiment group to further reflect on their emotions, the interface, system functions and the way of reply are all the same in both groups. Figure 2 - Right illustrates what the reflection questions and guiding prompts look like, we visualized the steps of self-awareness and converted them into a reflection form. The guiding prompts are highlighted in grey rows below (Table 2).

3.3 Participants Recruitment and Overview

We recruited a total of 20 participants (10 male, mean = 22.65, SD=1.24) via social media channels, and no incentives were provided (3 3). Participants were consented to participate in this study. Our recruitment inclusion criteria include those who have an IG

Table 2: Overview of reflection questions and guiding prompts for experiment group and control group

Guiding Prompts (G, for experiment group only) and Reflection Question (Q)		CG	EG
G1	Hi, how is today? Before we start, let's take 30 seconds to breathe deeply in a comfortable posture and at a speed you like to relax your body that allows your emotions to settle.		V
Q1	How long did you spend on Instagram today?	V	V
Q2	What did you do most when you use Instagram?	V	V
G2	Now, please imagine that you are an iceberg. Above sea level is the behavior you show today that is easy to be seen and understood. Below sea level is an emotion that is difficult to detect. You are invited to explore the appearance of the iceberg above and below sea level.		V
Q3	What is your mood at the moment?	V	V
Q4	Did this emotion bring you any physical feelings or changes?		V
G3	If you can dive deeper, maybe there are more stories of the iceberg that are worth exploring!		V
Q5	What's the reason or meaning behind the current emotional and physical feelings?		V
G4	Thank you for your willingness to see these feelings and reasons! Let's practice accepting these emotions and feelings together, and try to meditate in our hearts: "I am willing to be closer to this feeling, and I accept myself with these feelings."		V
Q6	How do you want to change or maintain your mood and state?		V
G5	We are about to end today's journey of exploring the iceberg. It's hard work! Don't forget: no matter what kind of story you explored, it's a deep footprint of your serious living in the present!		V
Q7	Now, you can give yourself some blessings or encouragement.		V

Table 3: Overview of participants characteristics

Control Group				Experiment Group			
ID	Gender	Age	FoMO Scale	ID	Gender	Age	FoMO Scale
C1	F	22	4.1	E1	F	23	2.5
C2	F	26	3.4	E2	F	22	4.0
C3	M	22	3.3	E3	M	22	3.5
C4	M	22	3.2	E4	M	22	3.3
C5	F	22	3.2	E5	F	23	3.2
C6	M	24	3.0	E6	M	23	3.0
C7	F	21	2.8	E7	F	22	2.9
C8	M	21	2.6	E8	F	22	2.7
C9	F	22	2.6	E9	M	25	2.7
C10	M	23	2.1	E10	M	24	2.3

account and experiences of FoMO on IG. We selected participants whose FoMO scores are from 2.1 to 4.1 on a total score of 5 through a pre-screening survey during the recruitment process. All participants self-identified themselves having the FoMO issue.

3.4 Measurements

Our study aims to examine how reflection questions can reduce negative social media effect, and how guiding prompts along with reflection questions could enhance people's emotion awareness. We capture participants' changes before and after the study by using existing survey instruments measuring people's perceptions of FoMO, emotion regulation, and emotional intelligence. FoMO scale [8] is a 10-item scale measuring people's statement of FoMO. The CERQ-short survey is a 18-item cognitive emotion regulation questionnaire developed by 10 to assess participants' emotion regulation [9, 10]. Finally, the Wong and Law Emotional Intelligence Scale (WLEIS) is used to measure individuals' emotional intelligence in

a total of four dimensions (self-emotion appraisal, use of emotion, regulation of emotion, and others' emotion appraisal) [17, 35]. This scale is composed of 16 items.

3.5 Data Collection and Analysis

We conducted the Wilcoxon sign rank test to analyze our quantitative data using SPSS. Our *Being* app collected log data on participants daily entries of their IG use behaviors and emotions after IG use, as well as their app use statistics (e.g., time stamps of daily entries, the number of characters of participant entries). We analyzed these data along with our interview data manually to better understand participants' behavior patterns. Our interview protocol contains questions on study participation experience, reactions toward the reflection questions and the guiding prompts, reactions toward their self-exploration process of emotions. All interviews were recorded, transcribed verbatim, and analyzed through the affinity diagram [20] by the whole research team.

4 RESULTS

4.1 Overview of App Usage

All participants fully participated in our field experiment. Participants in both groups spent 154.55 seconds in the *Being* app, and wrote down 52.05 words on average per day. For the control group, the average app use time is 96.45 seconds, and participants wrote 33.33 words on average to answer the three reflection questions. As for participants in the experiment group who followed additional guiding prompts, they spent 212.65 seconds in the app, and wrote 70.76 words on average per day.

To examine whether participants' daily entries comply with the guidelines and definitions mentioned in the Satir model, we manually went through each response, and calculated the compliance rate (%) for each participant. Results suggested that participants in both groups had 83% compliance rate on average. Participants in the control group had a 88 % compliance rate while those in the experiment group had a 78% compliance rate.

4.2 Reactions to the Reflection Questions in the *Being* App

Reflection questions aim to enable users to notice their behavior and emotion. They show some positive effects and are perceived to change their behavior and awareness of emotions by using the *Being* app. Results are presented in more detail below.

4.2.1 Positive effects of recording using the *Being* app. Some participants (C9, E4, E8) pointed out that they felt more focused on what happened to them by recording their emotions and IG use habits in the app (*"I feel that making this record will allow me to focus more on what happened, and when looking at it, I will focus more on what is happening to me rather than what is happening to others. - C10"*).

4.2.2 Perceived change: reduced IG use time. According to the interview data, five interviewees indicated that they paid more attention to their behaviors (C5, C7, C9, E6, E10) of use time on IG, and this helped them reflect on their use habits. Participants also realized their IG use time was different from what they originally imagined (underestimate their use time) during study participation (C5, C10, E4). As a result, some participants (C5, C10, E4) reduced their use time, such as turning off or not using the mobile phone before going to bed (*"I found that my use time was so much higher than expected. So, I actively reduced the use time from the next day. - C10"*).

4.2.3 Perceived change: increased awareness of emotions. We also found that the *Being* app prompted some participants (C5, C7, E4, E6, E10) to pay attention to their feelings (*"After using the Being, it was obvious that I remember my current emotions, and found that I noticed my feelings more easily. - E4"*). Two interviewees (C7, E8) also recalled negative emotions and tried to express the feelings via *Being*. Furthermore, some participants (C5, C10, E5) mentioned that this experiment also helped them clarify or redefine the meaning of using IG, and reflected on how life is affected by the information shared in the online community. In addition, one participant (C5) indicated she was able to better handle her emotions. For example, when arguing with her family or friends, she can immediately know her mood and calm down before communicating with others.

Moreover, our quantitative data suggested that participants were able to refocus themselves on joyful and pleasant events instead of getting stuck in emotions from the unpleasant events by answering reflection questions daily throughout the experiment. This was reflected in the results of the Wilcoxon signed-rank test in that a significant effect was observed on positive refocusing (a sub-dimension of emotion regulation) after the experiment ($z = -2.437$, $p = .015$).

4.3 Reactions Toward the Guiding Prompts in the *Being* App

4.3.1 Effect on self-awareness and self-affirmation. Participants in the experiment group pointed out that the guiding prompts helped them examine their emotional states more deeply with the iceberg's concretized imagination (*"There are some emotions that I forgot at the moment, but through self-awareness, the emotions may impact me again - E4"*). Besides, some participants (E4, E5) pointed out writing blessings in the app (Q7) helped enhance self-affirmation and made them feel encouraged. It was especially helpful to reduce anxiety or negative emotions. One participant specifically mentioned that he wrote down the blessings to himself and they served as encouragement for himself in the future (*"I will predict what myself will be tomorrow, and use that situation as an encouragement, just like a feeling of encouragement for my future self. - E4"*).

4.3.2 Effect on emotion regulation. Based on the Mann-Whitney U test results, participants in the experiment group improved in the "other-blame" sub-dimension underneath emotion regulation after the study ($z = -2.595$, $p = .009$). In other words, they became less likely to blame others for their negative emotions using social media, as an example of an improvement in terms of emotion regulation.

5 DISCUSSION

5.1 Effectiveness of the Reflection Questions in the *Being* App

Both groups were able to properly complete the experiment, achieving above 75% compliance rates. Our first research question asks how emotion awareness-based reflection questions help reduce participants' negative emotions caused by using social media in both groups. Both our qualitative and quantitative data suggested that the majority of participants started caring more about how long they spent on IG and paid attention to their use trend across days. The reflection process made participants think further on how they used IG both physically (use behaviors) and mentally (emotions aroused during use). Moreover, six out of ten interviewees from both groups attempted to decrease their use time on IG (C5, C7, C9, C10, E6, E10). These participants realized they could use the time saved from using IG too long to complete other productive tasks to feel more fulfilled. Among these participants, four people indicated they started engaging themselves more wisely in social media (C5, C10, E6, E10). Looking from the perspective of content consumption on social media, five people from both groups said they have learned how to take what they saw on IG more neutrally and calmly after the study (C4, C5, C10, E4, E5).

Furthermore, prior study has discovered that high FoMO is related to poor emotion regulation [2]. Although there was no statistically significant change in participants' emotional intelligence scores (no change in terms of their emotion regulation capability), we observed that two participants perceived their emotion awareness improved, and that they got to manage their emotions better. Strategies mentioned by our participants echo with the emotion regulation strategies mentioned in [25], such as cooling down emotions before taking any actions.

5.2 Effectiveness of the Guiding Prompts in the *Being* App

Our second research question asks how the inclusion of guiding prompts on top of the reflection questions helped reduce the negative emotions caused by using social media for participants in the experiment group. The inclusion of the guiding prompts aims to engage participants in reflecting even more deeply on their emotions and feeling after social media use. To our surprise, our findings showed that the addition of guiding prompts delivered via the *Being* app did not bring about a statistically significant effect on participants' emotional awareness. According to the app log data, participants only responded to these prompts with simple phrases without carefully performing the reflection based on the guiding prompt content. Our interview data suggested that this could be due to the lack of interactivity, as the interactivity level is high when Satir's theory is originally applied in psychological counseling. In other words, participants indicated they preferred reading less content in the guiding prompts themselves as these are additions to the reflection questions. The reflection questions on emotion awareness were better perceived by participants in the experiment group because they were quick and concise. We believe future improvements include keeping the quick and concise reflection questions in the intervention, while making the guiding prompts more interactively or visually appealing (aesthetically appealing sketches/pictures) to better engage participants in thinking and sharing, similar to the therapy cards commonly used in psychological counseling sessions to facilitate sharing and communication [24].

5.3 Negative Emotions Aroused from Reflection

Two participants recalled some negative emotions (C7, E8) during study participation that made them become more aware of their negative emotions. They also realized they were not as emotionally aware as they thought they were. The potential negative emotions caused by using existing mental health applications remain a design challenge. Research has suggested that existing and future design could consider offering help when users encounter emergent situations [33].

6 LIMITATIONS

Our study has three primary limitations. First, the study would benefit from a larger sample size over a longer period of experiment time. Second, the reflection content in the *Being* app is primarily informed by the framework of Satir's Personal Iceberg Metaphor. We believe each framework or model related to self-awareness could bring about different benefits to users. Third, the current

Being app does not provide users customization features such as personalized meditation music. We expect to enhance the system functions in the future.

7 CONCLUSION AND FUTURE WORK

Our study aims to develop an emotion awareness intervention based on the Satir Model to help people become more aware of their emotions after using IG through a set of reflection questions and guiding prompts. We built a web-based application called *Being* to deliver these content in a 10-day field study with 20 participants. Our findings suggested that participants in both groups became more aware of their use habits and emotions when using social media. Participants who were exposed to the guiding prompts in addition to the reflection questions have developed some skills in emotion regulation and reducing social media use. Future work involves re-designing the guiding prompts in the *Being* app to incorporate more interactivity and developing interventions based on different self-awareness frameworks to determine which features are best for users.

ACKNOWLEDGMENTS

We thank the anonymous reviewers for their useful inputs as well as feedback from Patty Ko. Big thanks to our study participants for their participation and feedback. This work was supported by the Ministry of Science and Technology in Taiwan (MOST grant 109-2221-E-007-063-MY3).

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